

Test code MAC instruction

Format: ARIT RD, RS, RT; LW RT, INM(RS); SW RT, INM(RS); BEQ RS, RT

Data Memory: `/*Constants*/ 256, 1, 8, 0, 0x80000000, 0xAB0, 0x00000BAD, 0x0BAD0C0D, /*Weights Mem[8-11]:*/ 0x27CF25CC, 0xF8CBE20A, 0x2BE4DE0E, 0xEE2129C4, /*Activations Mem[12-15]:*/ 0x12E625CC, 0xD2D81DCE, 0xF6C9D62F, 0x2BD1FDD7, /*bias Mem[16]*/ 0x000004BB, /*expected output Mem[17]*/ 0x00002D42 ...], /*y Mem[18]*/0x00000000]`

Pseudo-código neuron_relu_16

Math formulation: $y = \max(0, \text{bias} + a_0w_0 + a_1w_1 + \dots + a_{15}w_{15})$

// Neuron parameters

`/* Fixed-point format:`

`* activations, weights: int8 Q4.4 (x / 16)`

`* bias, output: int32 Q24.8 (x / 256)`

`*/`

`static const int8_t weights[16] = { 39, -49, 37, -52, -8, -53, -30, 10, 43, -28, -34, 14, -18, 33, 41, -60 };`

`/* real = [ 2.4375, -3.0625, 2.3125, -3.2500, -0.5000, -3.3125, -1.8750, 0.6250, 2.6875, -1.7500, -2.1250, 0.8750, -1.1250, 2.0625, 2.5625, -3.7500 ] */`

`static const int32_t bias = 1211; /* real = 4.73828125 */`

// INPUTS

`int8_t activations[16] = { 18, -26, 37, -52, -46, -40, 29, -50, -10, -55, -42, 47, 43, -47, -3, -41 };`

`/* real = [ 1.1250, -1.6250, 2.3125, -3.2500, -2.8750, -2.5000, 1.8125, -3.1250, -0.6250, -3.4375, -2.6250, 2.9375, 2.6875, -2.9375, -0.1875, -2.5625 ] */`

// Expected output:

`static const int32_t expected_output = 11586 (0x2D42); /* real = 45.2578125 */`

`int32_t y = bias ;`

`for (int i = 0; i < 16; i++) {y += (int32_t)activations[i] * (int32_t)weights[i];}`

// ReLU

`if (y < 0) {y = 0;}`

TAG	@(hex)	Inst(hex)	Instructions	Comments
Reset	0	10210003	beq R1, R1, INI;	Jump to INI (@16)
IRQ	4	00000000	NOP	IRQ not supported
DAbsort	8	00000000	NOP	Abort not supported
UNDEF	C	00000000	NOP	Undef not supported
INI:	10	081F0000	LW R31, x0(R0)	R31=Mem(0) = 256; R31 is the stack pointer (SP)
Main:	14	08010020	LW R1, x20(R0)	R1=Mem(32)=W[0-3]; memory word=32/4=8
	18	08020024	LW R2, x24(R0)	R2=Mem(36) =W[4-7];
	1C	08030028	LW R3, x28(R0)	R3=Mem(40)=W[8-11];
	20	0804002C	LW R4, x2C(R0)	R4=Mem(44) =W[12-15];
	24	08050030	LW R5, x30(R0)	R5=Mem(48)=A[0-3];
	28	08060034	LW R6, x34(R0)	R6=Mem(52) =A[4-7];
	2C	08070038	LW R7, x38(R0)	R7=Mem(56)=A[8-11];
	30	0808003C	LW R8, x3C(R0)	R8=Mem(60) =A[12-15];
	34	04255005	MAC_ini R10, R1 ,R5	R10, ACC = MAC(R1,R5);
	38	04465004	MAC R10, R2 ,R6	R10, ACC = MAC(R2,R6)+ ACC;
	3C	04675004	MAC R10, R3 ,R7	R10, ACC = MAC(R3,R7)+ ACC;
	40	04885004	MAC R10, R4 ,R8	R10, ACC = MAC(R4,R8)+ ACC;
	44	08090040	LW R9, x40(R0)	R9 = bias = Mem(64)
	48	00000000	NOP	
	4C	00000000	NOP	
	50	052A5000	ADD R10,R9,R10	R10 = R10 + bias;
Relu:	54	080B0010	LW R11, x10(R0)	R11 = 0xEFFF; sign mask
	58	00000000	NOP	
	5C	00000000	NOP	
	60	056A5802	AND R11, R10, R11	R11 = signbit&31 0s
	64	00000000	NOP	
	68	00000000	NOP	
	6C	11600002	BEQ R11, R0, Test	Brach if positive
	70	00000000	NOP	
	74	05405002	AND R10, R10, R0	R10 = 0; RELU op: If negative, set to 0
Test:	78	080C0044	LW R12, x44(R0)	R12 = expected result = Mem(68)
	7C	00000000	NOP	
	80	00000000	NOP	
	84	114C0005	BEQ R10,R12, END	
	88	00000000	NOP	
Error:	8C	08010018	LW R1, x18(R0)	R1=Mem(24) = 0x00000BAD;
	90	00000000	NOP	
	94	00000000	NOP	
	98	0C017004	SW r1, x7004(r0)	IO output <= 0x00000BAD;
Err_loop	9C	1000FFFF	BEQ R0,R0, Err_loop	Infinite loop. Wrong result
	A0	00000000	NOP	
END:	A4	0C0A0048	SW R10, x48(R0)	Mem(72) = R10
	A8	0C0A7004	SW r10, x7004(r0)	IO output <= 0x00001B35;
End_loop	AC	1000FFFF	BEQ R0,R0, End_loop	Infinite loop. Right result
	B0	00000000	NOP	